



**BHASKARACHARYA NATIONAL INSTITUTE FOR
SPACE APPLICATIONS AND GEO-INFORMATICS**
WEEKLY PROGRESS REPORT (25/05/2026 – 31/05/2026)

WEEK 2

PROJECT NAME	Prithvi EO Models (IBM Model)
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PROJECT DESCRIPTION :

Fine tuning and Training of IBM Prithvi EO Models for Earth Observation tasks.

GROUP MEMBER :

Dhanya R, CH Siddartha

GROUP ID :

2026G264

GROUP GUIDE :

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COLLEGE NAME :

Gati Shakti Vishwavidyalaya, Vadodara

25/05/2026	- Understood the difference between classification and segmentation, implemented pixel-level segmentation using Prithvi-EO-300M with UperNetDecoder - Generated pixel-wise masks using NDVI, NDWI, and NDBI spectral indices, and prepared 19,553 labelled patch pairs for segmentation training.
26/05/2026	Trained the Prithvi-EO-300M segmentation model on real Sentinel-2 data, generated a full 10980×10980 pixel segmentation output GeoTIFF covering the Gujarat region with four land use classes — Vegetation, Urban, Water, and Barren.
27/05/2026	Validated the segmentation output in QGIS by overlaying it on Google Satellite basemap, converted the output to XYZ tiles using QGIS, and built an interactive browser-based map viewer using Leaflet with opacity control and legend.
28/05/2026	Holiday (Bakrid)
29/05/2026	Installed and configured GeoServer, published the segmentation output as a geospatial layer, created and applied an SLD style for the four classes.
30/05/2026	- successfully tested all four OGC standard services — WMS, WFS, WCS, and WMTS - Built image pyramids using GDAL with nearest neighbour resampling for optimized multi-scale rendering.
31/05/2026	Holiday (Sunday)

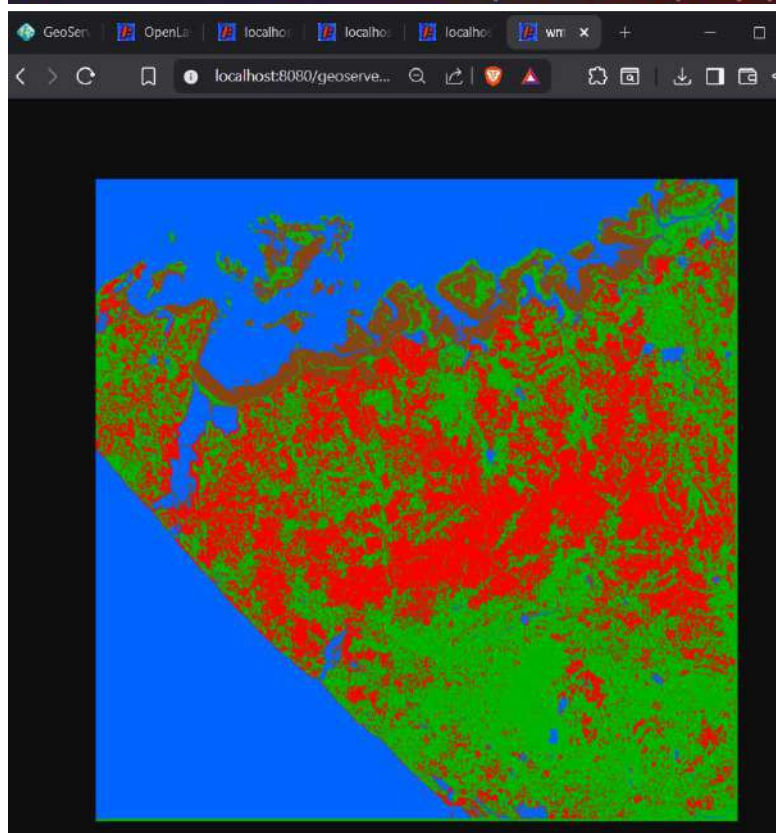
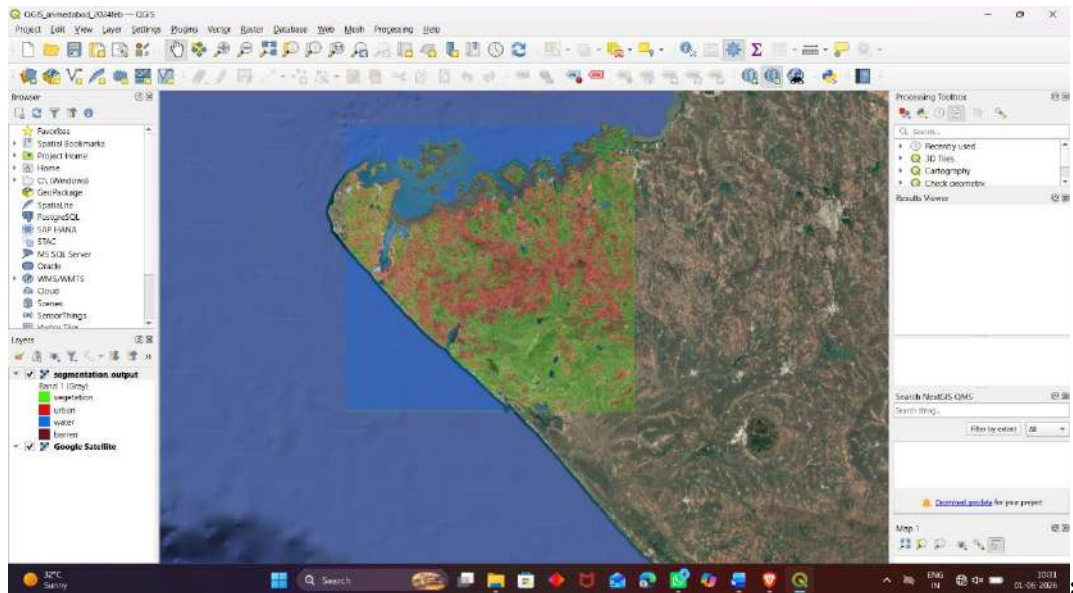
Next Week Plan	Planning to develop a Streamlit-based end-to-end application integrating the Prithvi-300M model with GeoServer REST API and a COG database..
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REFERENCE:

- [ibm-nasa-geospatial \(IBM-NASA Prithvi Models Family\)](#)
- [Sentinel-2 | Copernicus Data Space Ecosystem](#)
- <https://github.com/NASA-IMPACT/Prithvi-EO-2.0>

- <https://qgis.org/>
- <https://geoserver.org/>

SCREEN SHOTS



Business 27088+ 2540+ GitHub Workbooks Blotter Printful Spatial Coperni Segmen Explains Workitem GeoSer Geo x +

localhost:8080/geoserver/web/30

GeoServer

Logged in as admin

Welcome

GeoServer Web Service admin access to 3 workspaces, with 1 layers.

For more information visit [30\(organization\)](#).

1 Layers	1 Layer groups	1 Styles	1 Workspaces
Add new layer	Add new layer group	Add new style	Add new workspace

The embedded data directory is being used. It is **highly** recommended that you change it to a location outside of the web application.

The default user/group service should use digest password encoding.

The administrator password for the server has not been changed from the default. It is **highly** recommended that you change it now. [Change it](#)

Strong cryptography available

WMS

WMS	WMS
1.3.0	1.1.1

GeoServer Web Map Tile Service

Predefined map tiles generated from vector data, raster data, and images.

WMTS	WFS	WFS-C
1.0.0	1.0.0	1.1.2

WFS

WFS	WFS	WFS
2.0.0	1.1.0	1.0.0

WCS

WCS	WCS	WCS	WCS	WCS
2.6.1	1.1.1	1.1.0	1.0	1.0.0

Admin & Status

- Server Status
- GeoServer Logs
- Geotiff Information
- Reveal GeoServer

Data

- Layer Preview
- Workspaces
- Styles
- Layers
- Layer Groups
- Store

Services

- WMTS
- WCS
- WFS
- WMS

Settings

- Global
- Image Processing
- Raster Access

Tile Caching

- Tile Layers
- Caching Defaults
- Gridsets
- Disk Quota
- Statistics

Security

- Settings
- Authentication
- Authorization
- Users, Groups, Roles
- Data
- URL Checks
- Content Security Policy
- Services

Donor

Tools

Search

ENG IN 10:40 01-06-2025